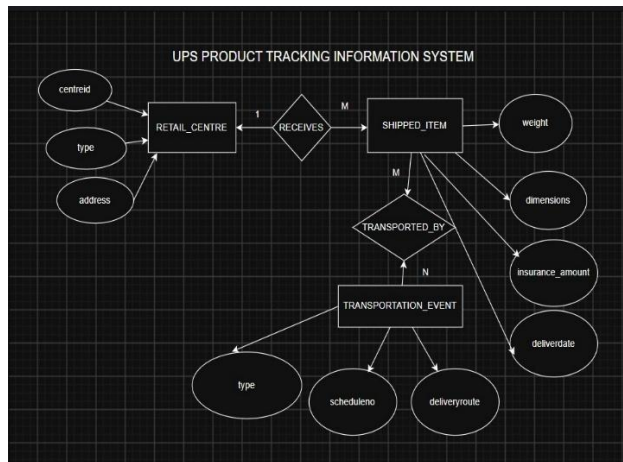


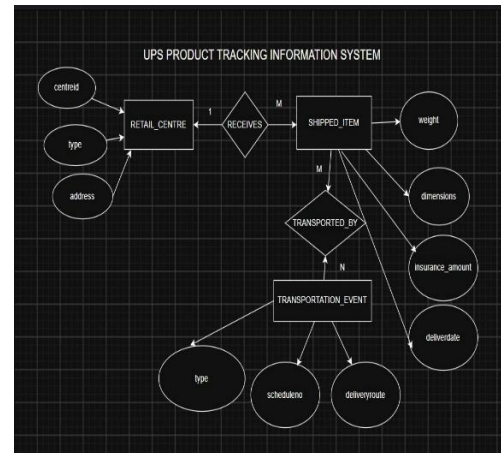
CO1 ASSESSMENT TOOL 1

NAME:AAMEENA I(192511284)

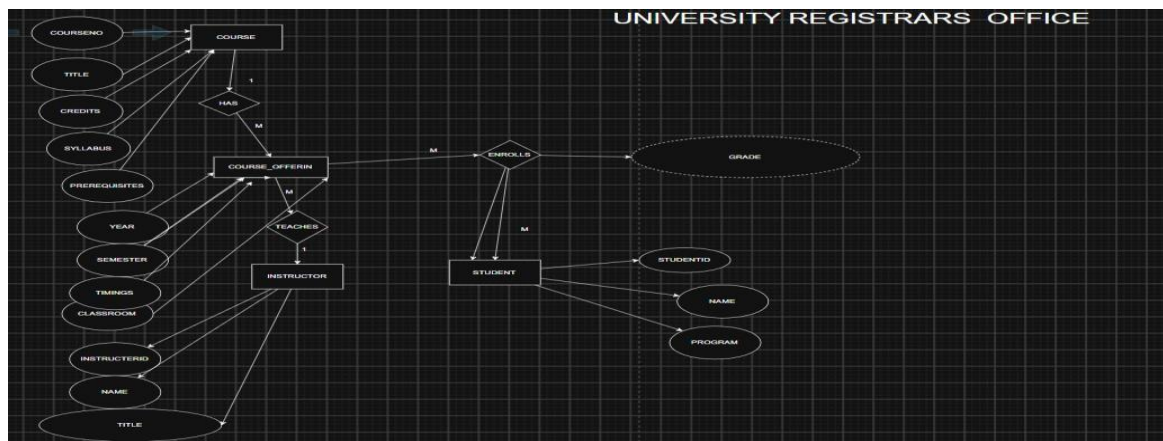
Q1)



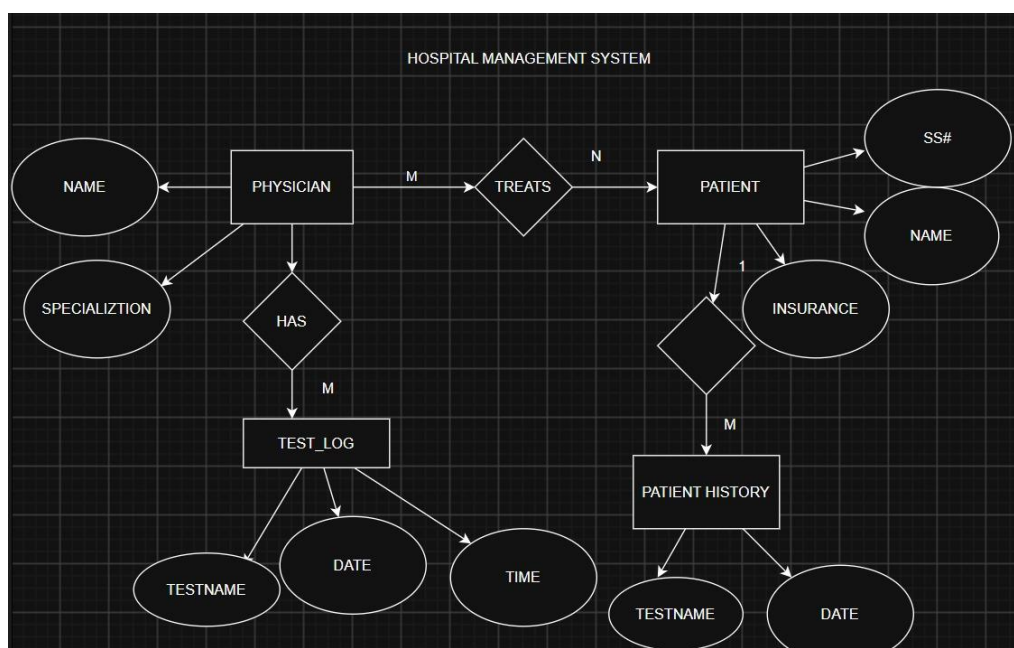
Q2)



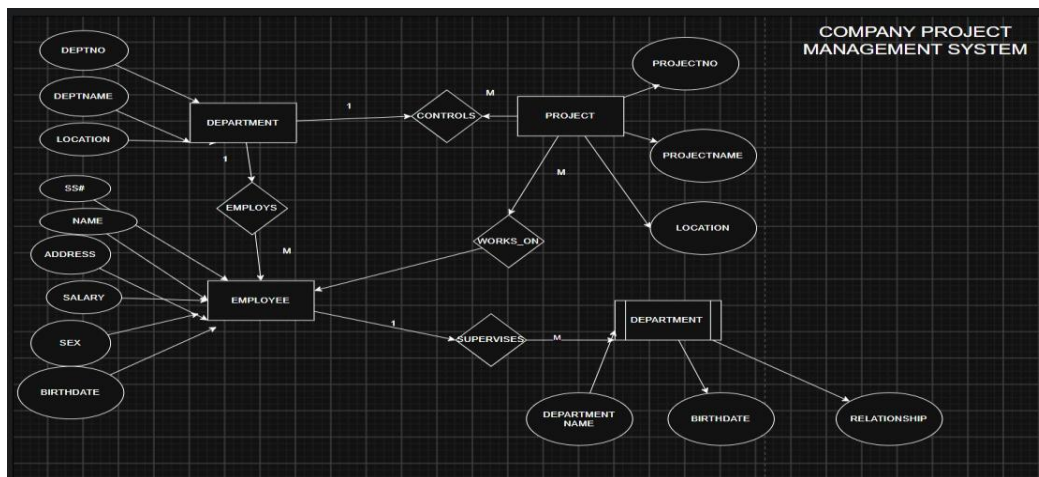
Q3)



Q4)



Q5)



Q6) Schema and Instance

1. Schema

A schema is the logical design or structure of a database. It defines the organization of data, including tables, attributes, relationships, and constraints. A schema is created during database design and remains stable for a long period.

Example:

STUDENT Schema

STUDENT

Student_ID

Name

Department

Age

The above schema defines the structure of the STUDENT table but does not contain actual records.

2. Instance

An instance is the actual data stored in a database at a particular point in time. It changes frequently whenever data is inserted, updated, or deleted.

Example:

STUDENT Instance

Student_ID	Name	Department	Age
------------	------	------------	-----

101	Ravi	CSE	20
102	Priya	ECE	21
103	Arun	IT	19

The above records represent an instance of the STUDENT schema.

Types of Schemas

1. Logical Schema (Conceptual Schema)

- Logical schema describes the complete logical structure of the database.
- It defines tables, attributes, relationships, and constraints.
- It does not describe physical storage details.

Example:

STUDENT(Student_ID, Name, Department)

COURSE(Course_ID, Course_Name)

2. Physical Schema (Internal Schema)

- Physical schema describes how data is stored in the storage device.
- It includes file structures, indexes, and storage techniques.
- It is concerned with database implementation.

Example:

Student data stored in:

- Data files
- Index files
- B+ Tree index on Student_ID

3. View Schema (External Schema)

- View schema describes the way users see the database.
- Different users can have different views of the same database.

- It provides security by showing only required information.

Example:

Student View:

Student_ID

Name

Department

Faculty View:

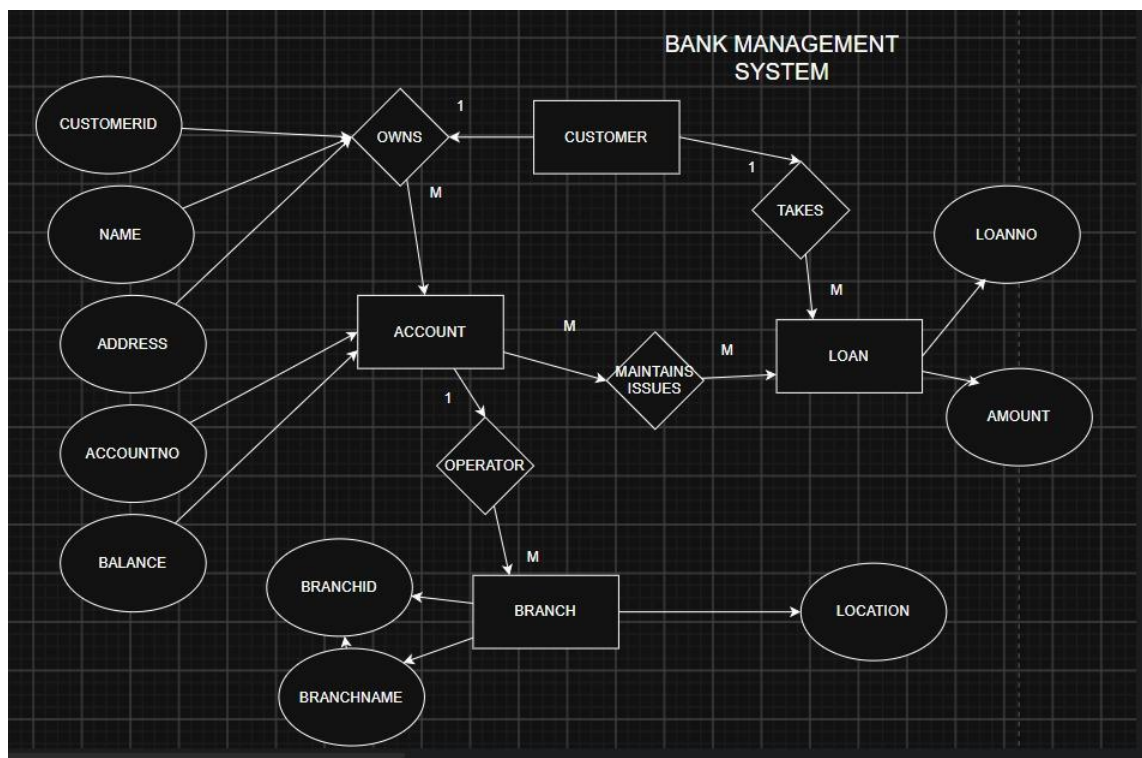
Student_ID

Name

Marks

Attendance

Q7)



Q8) Extended Entity Relationship (EER) Model

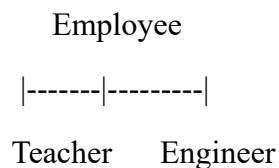
The Extended Entity Relationship (EER) model is an enhanced version of the ER model that provides additional concepts for representing complex real-world data. It includes all ER model concepts such as entities, attributes, relationships, and adds concepts like specialization, generalization, and aggregation.

The EER model is mainly used in database design to represent relationships between different types of entities more clearly.

1. Generalization

- Generalization is a bottom-up approach in which two or more lower-level entities are combined to form a higher-level entity.
- It identifies common attributes and relationships among entities and creates a generalized entity.
- The higher-level entity is called a superclass, and lower-level entities are called subclasses.

Example:



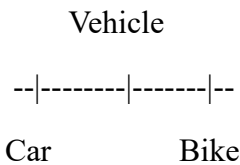
Explanation:

- Teacher and Engineer share common attributes such as Employee_ID, Name, and Salary.
- These common attributes are moved to the higher-level entity Employee.

2. Specialization

- Specialization is a top-down approach in which a higher-level entity is divided into lower-level entities based on specific characteristics.
- The higher-level entity is called a superclass, and the new entities are called subclasses.
- Each subclass inherits the attributes of the superclass.

Example:



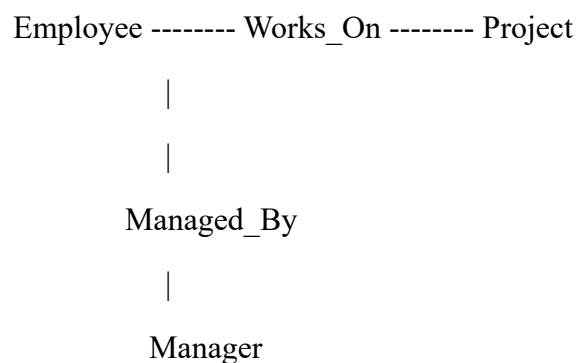
Explanation:

- Vehicle contains common attributes like Vehicle_ID and Model.
- Car and Bike have their own specific attributes.

3. Aggregation

- Aggregation is a concept used to treat a relationship set as a higher-level entity.
- It allows a relationship between an entity and another relationship.
- It is used when a relationship itself participates in another relationship.

Example:



Explanation:

- The relationship Works_On between Employee and Project is treated as an entity.
- A Manager can manage the Employee-Project relationship.

Advantages of EER Model

- Represents complex relationships effectively.
- Supports inheritance through superclass and subclass.
- Reduces data redundancy.
- Provides better database design.
- Improves understanding of real-world entities.

EER Model Structure

EER MODEL

Generalization Specialization Aggregation

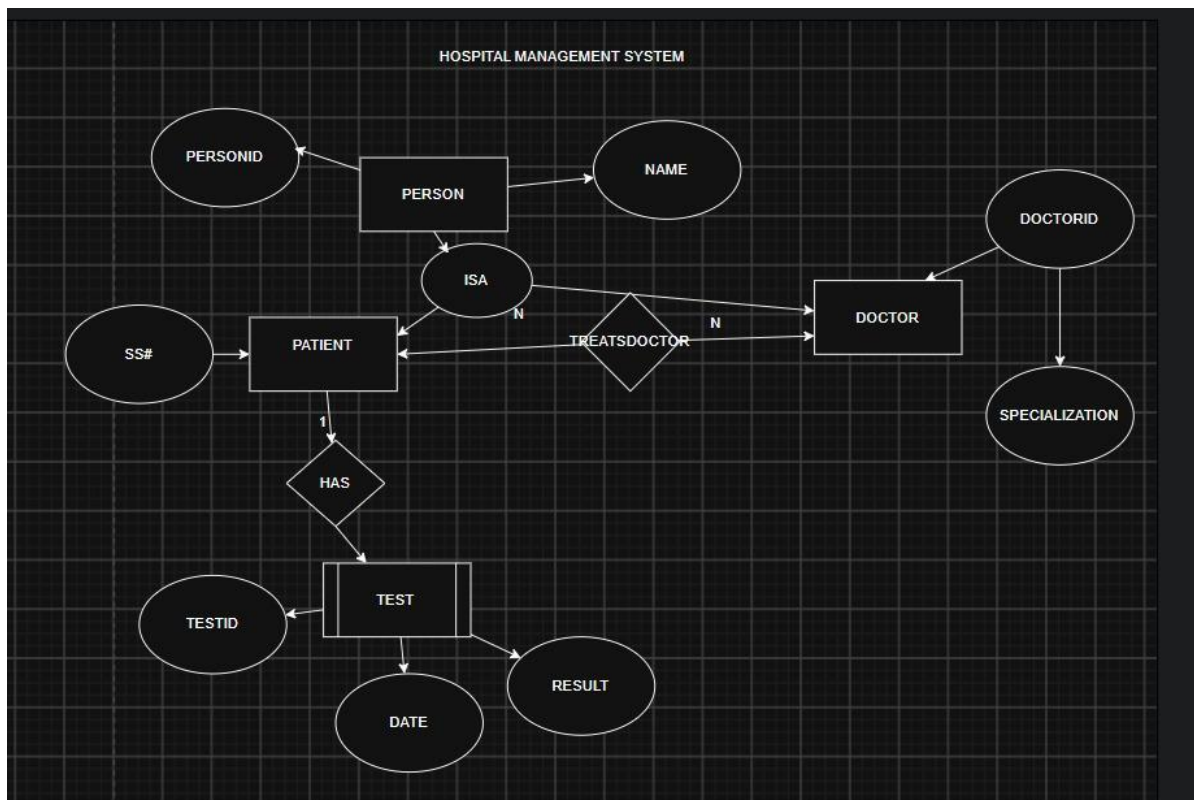
Combine Divide Relationship

Entities Entities as Entity

Conclusion:

The EER model extends the ER model by adding advanced concepts such as generalization, specialization, and aggregation. These concepts help designers create accurate and efficient database models for complex applications.

Q9)



Q10)Database Administrator (DBA)

A Database Administrator (DBA) is a person responsible for managing, maintaining, and controlling the database system of an organization. The DBA ensures that the database is secure, available, efficient, and performs properly.

The DBA acts as the administrator of the database environment and manages users, data storage, security, and backup operations.

Responsibilities of a DBA

1. Database Design

- Designs the database structure according to organizational requirements.
- Defines tables, relationships, constraints, and data models.

2. User Management

- Creates and manages database user accounts.
- Assigns roles and permissions to users.

3. Security Management

- Protects the database from unauthorized access.
- Implements authentication and access control mechanisms.

4. Backup and Recovery

- Performs regular database backups.
- Restores data during system failures or data loss.

5. Performance Tuning

- Monitors database performance.
- Optimizes queries, indexes, and storage methods for faster access.

6. Authorization and Access Control

- Grants or removes access privileges.
- Ensures users can access only authorized data.

7. Data Integrity Management

- Maintains accuracy and consistency of data.
- Enforces integrity constraints and validation rules.

8. Database Maintenance

- Updates database software and performs routine maintenance.
- Monitors database health and resolves issues.

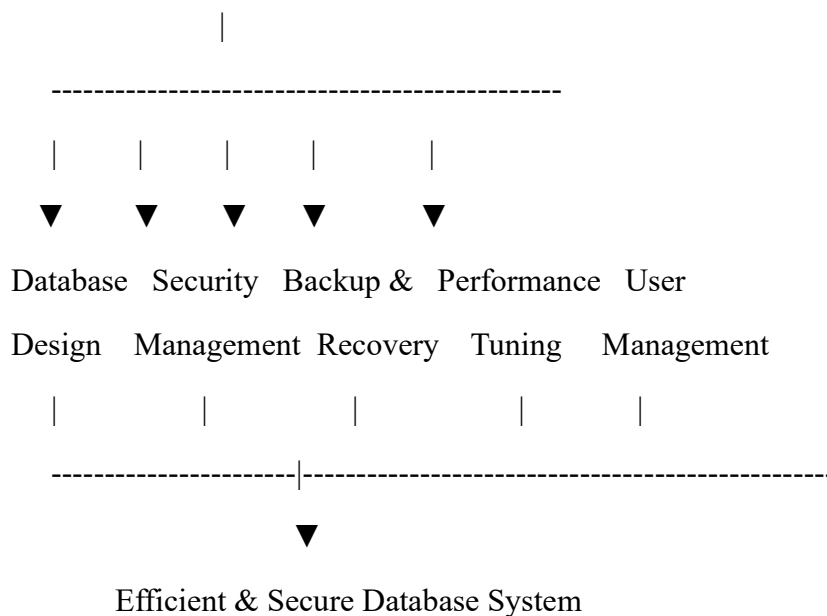
9. Storage Management

- Manages physical storage space.
- Organizes database files and storage structures.

10. Monitoring and Troubleshooting

- Detects and fixes database errors.
- Monitors system activities and resolves performance problems.

DATABASE ADMINISTRATOR (DBA)



Conclusion: The DBA plays a critical role in managing the database environment. By handling security, backup, performance, and user access, the DBA ensures that the database remains secure, reliable, and available for users and applications.

